SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
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Course Code: **DSA0613** **Slot: A**

Course Name: **Data Handling and Visualization for Data Analytics**

Course Faculty: DR. KUMARAGURUBARAN T & DR. SENTHILVADIVU S

Project Title: Healthcare Data Analysis and Visualization for Disease Trend Monitoring Using R

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)

Project Description: This module focuses on processing the acquired heart rate data to remove noise, enhance signal quality, and extract meaningful health information. The digital heartbeat data received from the acquisition module is analyzed to detect normal and abnormal heart rate patterns. The processed data is then prepared for visualization and long-term monitoring, enabling effective health assessment and decision support.

Module 1: HEALTHCARE DATA COLLECTION AND PREPARATION

Information:

The Healthcare Data Collection and Preparation module involves importing healthcare datasets from CSV files and performing essential preprocessing tasks. The dataset includes patient demographics, medical conditions, admission dates, regional information, and billing details. Data cleaning techniques such as handling missing values, correcting data formats, and removing inconsistencies are applied. Time-based attributes such as year and month are derived from admission dates to support trend analysis. Each patient admission record is treated as one disease case, allowing aggregation and statistical analysis. The processed dataset is organized into a structured format that ensures accuracy and consistency for subsequent analytical modules.

Outcome:

The module successfully converts analytical results into clear and interactive visual insights. It enables effective communication of disease trends, regional impact, and key healthcare patterns, supporting informed decision-making, improved disease monitoring, and enhanced understanding of public health data.

Student Signature **Guide Signature**